PHISHING DETECTION & AWARENESS REPORT

Future Interns — Cyber Security Task 2

Assessment	Phishing email analysis
Sample	Microsoft impersonation email — Sample 01
Assessment type	Security education and awareness analysis
Classification	PHISHING

1. Executive Summary

This report documents the analysis of one phishing email that impersonates Microsoft and presents itself as an account-security notification. The investigation reviewed the message content, sender information, available email headers, authentication results, and the URL presented to the recipient.

Multiple indicators were identified, including a suspicious sender domain, failed SPF/DKIM/DMARC checks, urgent account-security language, impersonation, and a shortened URL. Based on the available evidence, the message is classified as **Phishing**.

The suspicious link was not opened and no credentials or other sensitive information were submitted during the analysis.

2. Objective & Scope

The objective of this task is to analyze phishing email indicators, classify the email risk, explain the attack in simple language, and provide practical awareness guidance that users can apply when handling suspicious messages.

- Analyze the supplied phishing email sample.
- Identify phishing indicators in the sender, headers, message content, and URL.
- Review SPF, DKIM, and DMARC authentication results.
- Classify the email as Safe, Suspicious, or Phishing based on observed evidence.
- Provide prevention guidance and employee Do's and Don'ts.

Scope: analysis was limited to the supplied email sample and its available header/body information. No interaction with the suspicious destination was performed.

3. Email Sample Overview

Field	Observed value
Subject	[Action Required] Unusual sign-in activity on your account
From	"Microsoft Account Team" <noreply@microsoftonline-verify.com>
Recipient	jacobcookofficial@gmail.com
Date	Thu, 06 Feb 2026 17:14:18 +0000
Claimed organization	Microsoft

The message claims that unusual sign-in activity was detected on a Microsoft account and directs the recipient to review recent activity. The combination of an account-security theme and a call for immediate action is relevant to the phishing assessment.

4. Header & Authentication Analysis

Authentication results

Authentication check	Observed result	Interpretation
SPF	**FAIL**	The sending source did not pass the domain's SPF authorization check.
DKIM	**FAIL**	The message did not pass the available DKIM verification.
DMARC	**FAIL**	The message did not pass the available DMARC authentication check.

Sending information

- Sending IP observed in the received header: 178.238.225[.]91
- Return-Path: noreply@microsoftonline-verify[.]com
- From address: noreply@microsoftonline-verify[.]com

These header details provide evidence that the message did not successfully authenticate through the available email authentication checks.

5. Sender & Domain Analysis

The email uses the display name **"Microsoft Account Team"** and claims to represent Microsoft.

The sender address uses the domain **microsoftonline-verify[.]com**. In the context of a message claiming to be an official Microsoft security notification, this domain should be treated with caution. The authentication failures provide additional evidence that the message is not trustworthy.

Why this matters

- Display names can be chosen by senders and are not sufficient proof of identity.
- The complete sender address and domain should be checked.
- Authentication results provide additional technical evidence when assessing a message.

6. Suspicious URL Analysis

The message contains the following shortened URL:

https://bit[.]ly/3vF9xKz

A shortened URL can conceal the final destination from the recipient. In this message, the shortened link is combined with an urgent account-security request, making it an important phishing indicator.

Safety note: the link was not opened or followed during this analysis.

7. Phishing Indicators Identified

Indicator	Evidence / explanation
Impersonation	The message presents itself as a Microsoft account-security notification.
Suspicious sender domain	The sender uses microsoftonline-verify[.]com while claiming to represent Microsoft.
Authentication failures	SPF, DKIM, and DMARC all show FAIL in the supplied header evidence.
Urgency and fear	The message warns about unusual account activity and encourages immediate action.
Suspicious shortened URL	The message directs the recipient to a bit[.]ly link.
Immediate action request	The recipient is encouraged to review recent activity through the supplied link.

8. Risk Classification

Classification: PHISHING

The classification is based on the combined evidence rather than a single characteristic. The message combines Microsoft impersonation, a suspicious sender domain, failed email authentication checks, urgent account-security language, a shortened URL, and a request for immediate user action.

9. How the Attack Works

1	**Impersonation**	The sender presents the message as a Microsoft security notification.
2	**Create concern**	The message claims unusual sign-in activity has occurred.
3	**Create urgency**	The recipient is encouraged to take immediate action.
4	**Provide a link**	A shortened URL is supplied as the route to review activity.
5	**Potential credential theft**	A victim who follows a fraudulent destination could be asked for sensitive information.

This describes the likely social-engineering flow represented by the message. No interaction with the suspicious destination was performed.

10. Prevention & Awareness Guidelines

Recommended user actions

- Do not click the suspicious link.
- Do not enter passwords, OTPs, or other sensitive information through an email link.
- Do not reply to the message.
- Verify account activity through the organization's official website or application.
- Report the suspicious email to the appropriate security or IT team.
- Preserve the email as evidence if an investigation is required.

Employee Do's

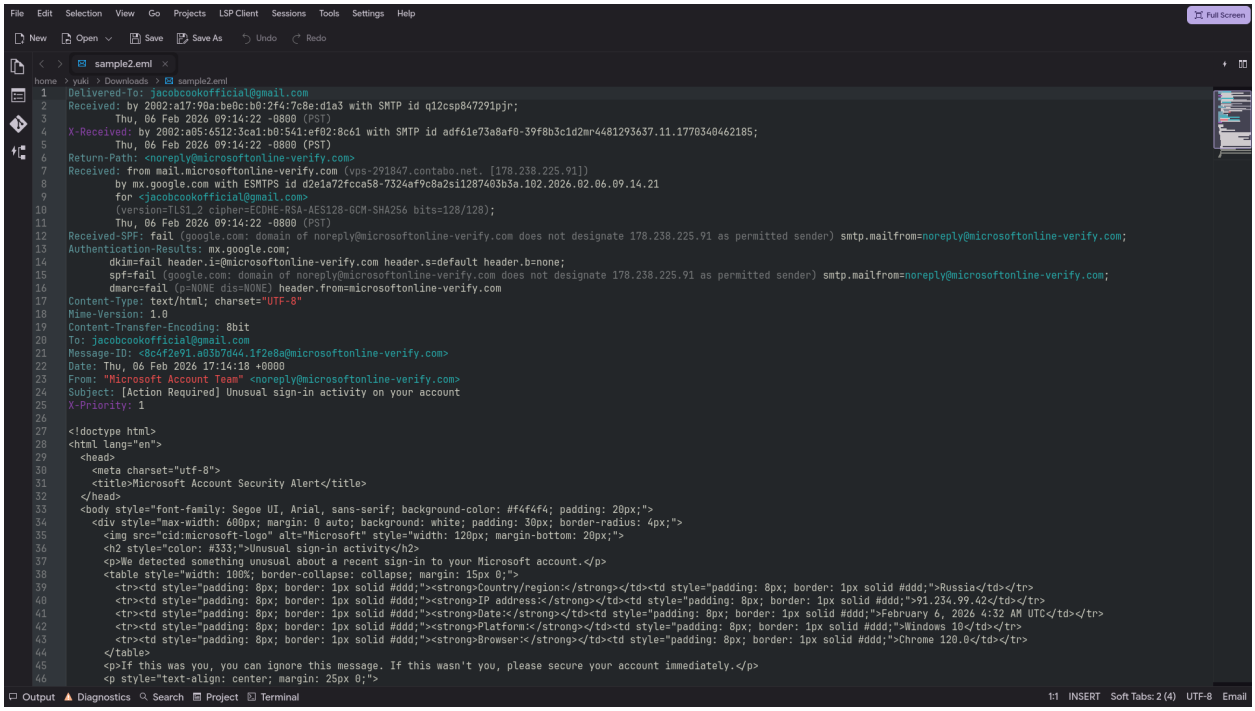
- Check the sender's complete email address.
- Be cautious with urgent security warnings.
- Verify account activity through an official website or application.
- Report suspicious emails.
- Ask the IT/security team when uncertain.

Employee Don'ts

- Don't click unexpected links.
- Don't provide passwords or OTPs through an email link.
- Don't trust an email simply because it uses a familiar company name or logo.
- Don't allow urgency to override normal security checks.
- Don't open suspicious links just to investigate them.

11. Evidence

The original email is preserved in the project repository as **sample/sample.eml**. A cleaned readable representation is stored as **sample/sample_email_readable.txt**. The following screenshot documents the header-level analysis performed on the original email.



Evidence screenshot: sample-email-analysis.png

12. Conclusion

The analyzed email contains multiple indicators associated with phishing, including impersonation, a suspicious sender domain, failed email authentication checks, urgent account-security language, and a shortened URL.

Based on the available evidence, the message is classified as **Phishing**.

The analysis was performed for cybersecurity education and awareness purposes. No credentials were submitted and the suspicious URL was not opened.

Repository evidence: sample.eml, sample_email_readable.txt, sample-email-analysis.png, and the supporting analysis document.